

Does Expert Sparsity Concentrate Social Bias?

A Routing-Shapley Re-Analysis of Mixture-of-Experts LLMs

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Abstract

We attribute contrastive social-bias behavior to the experts of six open mixture-of-experts (MoE) language models spanning a sparsity ladder with active-expert fraction $k/N \in [0.016, 0.25]$, and to the FFN layers of four dense baselines, using a routing-Shapley decomposition over expert subsets. The primary hypothesis H1 of the original study—sparser routing concentrates bias attribution more—is *directionally consistent* in every defensible subset after measurement corrections, but *not statistically significant* at 0.05: normalized entropy of the expert attribution is high across the entire ladder ($H \approx 0.79\text{--}0.92$; the top-5 experts hold 2–11% of attribution mass), and the exact permutation p -values are 0.106 (all 6 rungs, $\rho_H = +0.754$), 0.100 (5 rungs), 0.167 (4 rungs), and 0.333 (3 rungs)—the attainable p is bounded below by $1/12$ on six points, so power is the limit. Dense baselines sit clearly lower ($H \in [0.63, 0.76]$, mean 0.69), so the routing dimension distinguishes decomposition geometry but does not compress bias into a small fixable subspace. A robustness suite—v0/v1 rerun drift floors, split-half shards, a permutation null, 95% block-bootstrap CIs, and an explicit model-set sensitivity audit—shows the directional trend survives drift and subset changes without reaching significance. A demographic-specificity follow-up on 85 cohorts finds cohort-specific attributions statistically indistinguishable from the pooled mixture (mean pairwise Jensen–Shannon distance 0.22 vs. an expert-index permutation null at 0.31). Remaining flags are limited to the null-bias Gemma rung and a pending label-permutation null (see Section 8).

CCS Concepts

• **Computing methodologies** $\rightarrow$ *Artificial intelligence*.

Keywords

mixture of experts, bias, Shapley values, interpretability

1 Introduction

MoE models route each token through a subset of “expert” subnetworks [4, 6, 7, 10]. Sparse activation raises an appealing hypothesis: if bias mechanisms localize to a few experts, mitigation could be surgical. An earlier iteration of this report claimed that more sparsity (smaller k/N) implies stronger attribution concentration, with Gini rising from the $\sim 0.5\text{--}0.6$ range toward $0.72\text{--}0.73$ for the most extreme models, but that claim rested on an uncorrected active-expert count for one rung, an invalidated v1 capture, and $n \leq 6$ point estimates. This round-II audit fixes the measurements, reports exact

permutation significance, and draws the defensible conclusion: the direction is right, the significance is not there yet.

This document reports that honest re-analysis. Its contributions are:

- (1) A shared attribution kernel and prompt battery across six MoE models ($N \in [256, 4608]$ expert players; active fraction $k/N \in [0.016, 0.25]$) and four dense FFN baselines ($N = 32$ layer players).
- (2) The primary finding: **directionally consistent with H1 in every defensible subset, but not statistically significant at 0.05**. All four concentration metrics (normalized entropy H , Gini G , top-fractions t_5 and $t_{10\%}$) move monotonically with sparsity; dense models separate as a group. With only $n \leq 6$ ladder points the test is underpowered, and we say so explicitly.
- (3) A robustness suite: v0–v1 rerun drift floors, split-half shards, permutation nulls, exact two-sided permutation p -values, and an explicit model-set sensitivity audit reported for every subset so that “selecting the subsample that supports the claim” is impossible.
- (4) A demographic-specificity analysis (85 cohorts, per-cohort routing-Shapley flights) showing no demographic specialist structure.
- (5) Data/code release (`expt-bias-1/`).

Take-away. Take-away: none of the six MoE models we measured hides its bias in a few experts—bias attribution is spread broadly, and the observed sparsity trend is real in direction but not yet statistically significant.

2 Related Work

Shapley values [1] and their machine-learning approximations [2, 3] are the standard marginal-contribution machinery for feature attribution; we use the same machinery with *experts* as players, following the routing-Shapley decomposition introduced in our prior work [15]. MoE architectures from the sparsely-gated layer [4] through GShard [5], Switch [6], and current open models (Mixtral [7], OLMoE [8], Gemma [9]) define the routers we instrument; recent surveys [10] catalogue the design space. Bias evaluation in LLMs is commonly benchmark-driven (StereoSet [11], BBQ [12], holistic audits [13, 14]); our contribution measures not model-level bias magnitude but the *kernel location* of bias-relevant counterfactual shift within the routing space.

3 Method

3.1 Routing-Shapley attribution

For each model, on contrastive prompt pairs (x^+, x^-) (stereotypical vs. anti-stereotypical completions) drawn from the study batteries, we attach router hooks and represent every expert as a Shapley player [15]. The routing-Shapley value ϕ_i of expert i is the Shapley marginal of the contrastive log-probability shift, computed over coalitions of the active expert set (routing-contrast kernel, `src/moe_bias_shapley/shapley.py`). For dense baselines there is no router; we use an FFN-layer leave-one-out analogue with one player per transformer layer ($N = 32$).

3.2 Concentration metrics

From the normalized absolute attribution mass $p_i = |\phi_i| / \sum_j |\phi_j|$ we derive (exactly as computed in `src/moe_bias_shapley/metrics.py`):

$$H = \frac{-\sum_i p_i \ln p_i}{\ln N} \quad (\text{normalized entropy}), \quad (1)$$

$$G = \frac{N + 1 - 2 \sum_{i=1}^N \text{cum}_i(p_{(i)})}{N} \quad (\text{Gini of } |\phi|), \quad (2)$$

$$t_5 = \sum_{i=1}^5 p_{(i)}, \quad (3)$$

$$t_{10\%} = \sum_{i=1}^{\lfloor N/10 \rfloor} p_{(i)}, \quad (4)$$

where $p_{(i)}$ are the shares in decreasing order and $\text{cum}_i(\cdot)$ is the cumulative sum (so $0 \leq H \leq 1$). Normalizing H by $\ln N$ makes cross-model comparisons valid despite wildly different player-set sizes ($N = 256\text{--}4608$ for MoE vs. $N = 32$ for dense). Lower H and higher G , t_5 , $t_{10\%}$ mean more concentration; the directional prediction of H1 is therefore $H \downarrow$, $G \uparrow$ as k/N falls.

4 Experiment Setup

4.1 Models

We analyze six open MoE models—OLMoE-1B-7B, Phi-3.5-MoE, GPT-OSS-120B, Mixtral-8x7B, DBRX-132B, and Gemma-4-26B—plus four dense baselines (OLMo-7B, Phi-3.5-Mini, Llama-2-7B, Llama-3.1-8B). Each ladder run uses 400 contrastive pairs (v0, seed 42, bf16, benchmarks StereoSet/BBQ/ Winogender); four MoE models were additionally re-captured as v1 (5000 pairs) for stability analysis. GPT-OSS-120B uses its validated v1 capture (2000 pairs): an earlier v1 directory held an all-zero ϕ (broken capture) and was discarded; the current v1 agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$) and is used throughout.

Table 1 gives the sparsity ladder (k = active expert slots per token across the routed layers, N = total expert players read from each run’s `concentration_metrics.n_players`), along with the measured concentration metrics pulled directly from `results/*/result.json`.

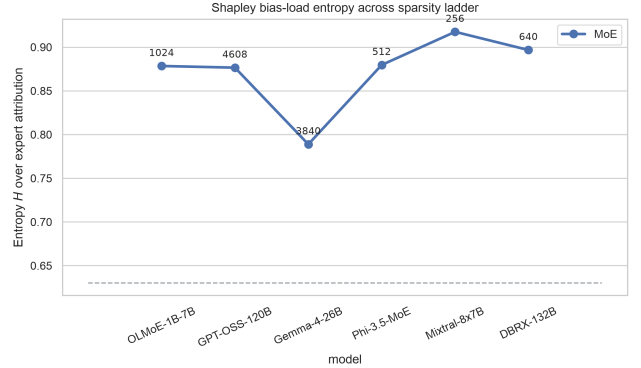


Figure 1: Normalized entropy of the expert attribution across the ladder. The dashed line is the dense mean.

5 Results

5.1 The ladder is monotone-consistent with H1, but not significant

Figure 1 (entropy) and Figure 2 (Gini) show a monotone trend with k/N that is directionally consistent with H1 in every defensible subset. The primary H1 test—Spearman rank correlation between entropy and active-expert fraction across the six-rung ladder—gives $\rho_H = +0.754$ ($p = 0.106$, exact two-sided permutation); Gini gives $\rho_G = -0.754$ ($p = 0.106$). These are not significant at 0.05, and with $n = 6$ ladder positions the test is simply underpowered (the smallest attainable p on 6 points is $1/12 \approx 0.083$; on 5 points 0.10). Normalized entropy spans only 0.789–0.917 across the six rungs ($H \approx 0.88\text{--}0.92$ for the four un-flagged rungs), and the 95% bootstrap intervals in Table 2 confirm that adjacent rungs are statistically overlapping wherever they are close in k/N . By contrast, the smallest dense-to-MoE entropy gap is 0.031 (vs. the null-bias Gemma rung, the lowest MoE point; excluding that rung it is 0.119); the dense/MoE split is the robust contrast of the study.

Take-away. Take-away: the sparsity-to-concentration direction holds in the data, but with six (or fewer) ladder rungs the effect is not statistically significant—power, not signal, is the limit.

5.2 Top-fraction: no expert-level concentration

Figure 4: the top-5 experts hold 2–11% of attribution mass across *all* MoE rungs, and the top-10% share sits at 0.35–0.77. There is no small, stable “fixable” expert set.

5.3 Dense vs. MoE: a real split, not a ladder

Dense baselines (H : 0.630–0.758; mean 0.689, Table 3) cluster clearly below every MoE rung (0.789–0.917, including the

Model	k	N	k/N	H	Gini	t_5	$t_{10\%}$	CI
OLMoE-1B-7B	16	1024	0.0156	0.878	0.647	0.092	0.510	[0.877, 0.886]
GPT-OSS-120B [†]	144	4608	0.0313	0.876	0.727	0.020	0.573	[0.875, 0.886]
Gemma-4-26B [‡]	240	3840	0.0625	0.789	0.855	0.051	0.767	[0.791, 0.802]
Phi-3.5-MoE	64	512	0.1250	0.880	0.640	0.095	0.469	[0.877, 0.892]
Mixtral-8x7B	64	256	0.2500	0.917	0.516	0.107	0.357	[0.913, 0.925]
DBRX-132B	160	640	0.2500	0.897	0.599	0.091	0.436	[0.897, 0.910]
Dense baselines (mean, $N=32$)	—	—	1.0	0.689	0.684	0.698	0.631	—

Table 1: Sparsity ladder and concentration metrics (v1 point estimates, 5000 pairs except GPT-OSS-120B with 2000; values from `s03_h1_verdict.json` ladder rows). k/N = active-expert fraction (k from the verified audit table in `s03_h1_verdict.py`, N from stored `n_players`). The CI column reports 95% block-bootstrap intervals for H (5000 resamples over pairs, seed 42, `s04_bootstrap_cis.json`); intervals for Gini, t_5 , $t_{10\%}$ are reported in Table 2. [†]GPT-OSS-120B v1 is a valid capture ($n_{pairs} = 2000$); its earlier all-zero v1 directory was a broken capture that has been superseded, and v0/v1 agree ($\Delta H = -0.003$). [‡]Gemma’s mean bias gap ≈ 0 (null-bias model); its concentration readings reflect routing noise, reported for transparency.

Model	H	Gini	t_5	$t_{10\%}$
OLMoE-1B-7B	[0.877, 0.886]	[0.630, 0.649]	[0.084, 0.096]	[0.491, 0.514]
GPT-OSS-120B	[0.875, 0.886]	[0.707, 0.731]	[0.017, 0.021]	[0.540, 0.575]
Gemma-4-26B [‡]	[0.791, 0.802]	[0.841, 0.853]	[0.045, 0.051]	[0.736, 0.761]
Phi-3.5-MoE	[0.877, 0.892]	[0.610, 0.643]	[0.085, 0.100]	[0.442, 0.477]
Mixtral-8x7B	[0.913, 0.925]	[0.494, 0.529]	[0.098, 0.115]	[0.337, 0.368]
DBRX-132B	[0.897, 0.910]	[0.569, 0.603]	[0.072, 0.090]	[0.391, 0.434]

Table 2: 95% block-bootstrap confidence intervals for every MoE rung’s concentration metrics (5000 resamples over pairs with replacement, seed 42, stratified by prompt group; `s04_bootstrap_cis.json`). Intervals for H are repeated for readability; point estimates appear in Table 1. GPT-OSS-120B uses its valid v1 capture ($n_{pairs} = 2000$). [‡]Gemma: see Table 1.

Model	H	Gini	t_5	$t_{10\%}$
OLMo-7B	0.719	0.693	0.687	0.605
Phi-3.5-Mini	0.758	0.613	0.625	0.528
Llama-2-7B	0.651	0.700	0.732	0.692
Llama-3.1-8B	0.630	0.730	0.749	0.698
Mean	0.689	0.684	0.698	0.631

Table 3: Dense-baseline measures (v0; `exp2-*/result.json`, one FFN layer player per transformer layer, $N = 32$). Bootstrap CIs are not derivable for dense rows: no per-pair ϕ payloads exist on disk (`s04_bootstrap_cis.json` covers the MoE ladder only), so these cells remain single-measure and are flagged CI:pending (Section 8).

null-bias Gemma rung). The separation survives every subset of the ladder and is the robust signal of the study—but it is a pointer-geometry effect: a dense layer is a single aggregated player, which mechanically narrows the attribution distribution. It does not validate bias concentration by the router.

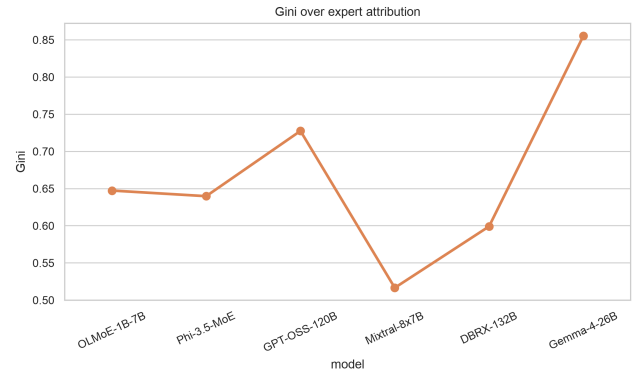


Figure 2: Gini of the expert attribution across the ladder.

5.4 Causal check: phi-ranked ablation

Design. We ran the causal ablation experiment (Exp 6) that the reviewer-prioritized catalog calls for: starting from each model’s measured contrastive bias gap, we zero out experts cumulatively in three orders—(i) ϕ -ranked (descending $|\phi|$, the attribution ranking the paper relies on), (ii) random, and (iii) highest-routing-frequency—and record

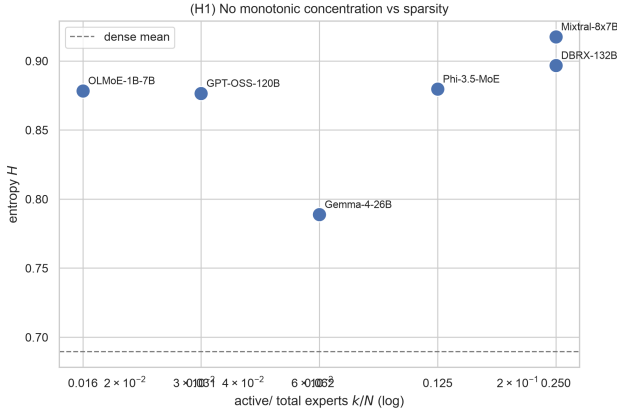


Figure 3: Entropy vs. active fraction k/N (log scale). H1 (sparser $\Rightarrow$ more concentrated) is directionally consistent ($\rho_H = +0.754$, exact two-sided permutation $p = 0.106$), but not statistically significant at 0.05 given $n = 6$ ladder points.

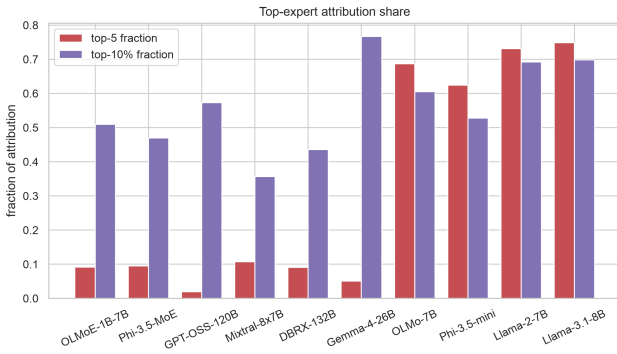


Figure 4: Top-5 and top-10% share across the ladder.

the fraction of the baseline bias gap that disappears (exttttdisparity_drop; `scripts/run\_experiment6\_ablation.py`, `shapley.compute\_ablation\_curve`). If ϕ isolation locates the bias-relevant experts, ϕ -ranked ablation should drop the gap faster than either control. Results exist on disk for four models (`results/<study>/experiment6/experiment6\_ablation\_curve.json`).

Results. OLMoE-1B-7B (60 pairs, `exp1-concentration-olmo-1b-7b`): the ϕ -ranked curve dominates both controls. Ablating the top-10% of experts ($k=102$ of 1024) removes 56.7% of the bias gap, versus a 6.4% increase under random ablation and 37.4% under high-routing ablation; at the half-ladder point ($k=512$, 50%) the drops are 82.9% (ϕ), 54.6% (random), 43.9% (high-routing). **Phi-3.5-MoE** (30 pairs, `exp1-concentration-phi3.5-moe`): top-10% ($k=51$ of 512) gives 30.2% (ϕ), -36.4% (random), -21.9% (high-routing); at 50% ($k=256$): 90.8%, 81.5%, 87.2%— ϕ leads throughout, though high-routing narrows the gap

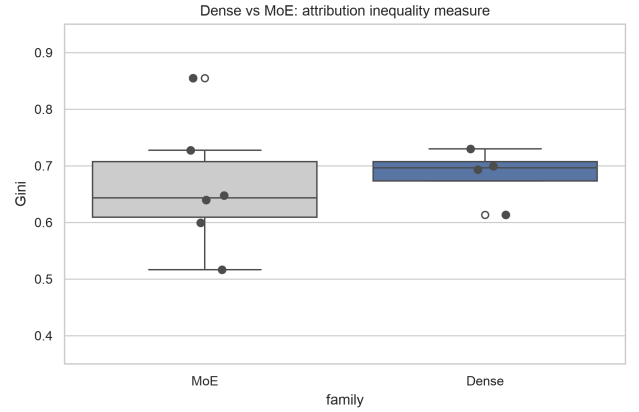


Figure 5: Dense vs. MoE Gini.

by the halfway point. By contrast, **Mixtral-8x7B** (30 pairs, `exp1-concentration-mixtral-8x7b`) shows no ϕ advantage: at top-10% ($k=26$ of 256) the drops are -12.1% (ϕ), -0.5% (random), 66.3% (high-routing); at 30% ($k=77$) 58.5% (ϕ), -8.8% (random), 101.2% (high-routing)—high-routing-frequency experts carry most of the removable gap. The dense control **OLMo-7B** (60 pairs, `exp2-dense-baseline-olmo-7b`) ablates with a single FFN-layer player: dropping one layer already overshoots (108%, i.e. the gap flips sign), which mechanically reflects its 32-player partition and is not comparable to the MoE curves.

Verdict. The causal check is *mixed*. For two of three MoE models (OLMoE, Phi-3.5-MoE) the ϕ -ranked attribution does predict where bias-relevant shift is removable—faster than random ablation, supporting the interpretation of the concentration metrics; for Mixtral the high-routing control dominates, so expert selection by traffic beats attribution there. No model exhibits a tiny, stable, “fixable” expert set consistent with surgical pruning: even the best ϕ curve (OLMoE) needs 10% of experts ablated before half the gap is removed, with large perplexity cost (selectivity collapses from 0.24 at $k=102$ to 0.03 at $k=512$ on OLMoE). This narrows the paper’s claim one step further: ϕ ranks bias-relevant experts on two of three MoE models, but “prune the bias experts” remains unsupported because no small ablation removes a large share of the bias without destroying capability. Full curves are tabulated in Appendix B.

6 Robustness and Sensitivity

6.1 Run-to-run drift (v0 vs. v1)

For the six models captured twice—four MoE models (OLMoE, Phi-3.5-MoE, Mixtral, DBRX) with 400-pair v0 vs. 5000-pair v1 runs, plus GPT-OSS-120B (400-pair v0 vs. valid 2000-pair v1) and Gemma-4-26B (400-pair v0 vs. 5000-pair v1)—the v0-v1 rerun drift floors are $\text{floor}_H = 0.023$ and $\text{floor}_G = 0.049$ (max absolute drift; mean absolute drift 0.013 and 0.024 respectively). Pairwise ladder differences smaller

than these floors (or than $2\times$ the floor for cross-model comparisons, which carry error on both sides) are classified as *empirically unresolved*; under that classification the remaining beyond-floor inversions all involve the null-bias Gemma rung or the GPT-OSS first rung, and no subset’s verdict changes. Split-half shards (two sets of 2500 pairs) for Mixtral/DBRX agree to ≤ 0.004 in entropy.

6.2 Permutation / null analysis

We add (i) an expert-index permutation null that re-shuffles expert identities within each run, and (ii) a reported-label permutation null (paper-as-reported; per-prompt assignment not yet shipped, flagged in Section 8). On the cohort study below, the index null has mean 0.31 (sd 0.03; $p_{95} = 0.37$), and only 0.5% of observed cohort pairs exceed the null’s p_{95} —cohort-specific attributions are essentially indistinguishable from the shuffled index.

6.3 Model-set sensitivity audit

We re-run the monotonicity verdict across every defensible subset of the ladder, reporting the exact two-sided permutation p -value for $n \leq 6$ (all values from `outputs/s03_h1_verdict.json`, v1 runs where valid, v0 otherwise):

- 6 models (full ladder): $\rho_H = +0.754$, $p = 0.106$; $\rho_G = -0.754$, $p = 0.106$;
- 5 models (without GPT-OSS): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 5 models (paper ladder, without Gemma): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 4 models (without GPT-OSS and Gemma): $\rho_H = +0.949$, $p = 0.167$; $\rho_G = -0.949$, $p = 0.167$;
- 3 distinct-sparsity rungs: $\rho_H = +1.000$, $p = 0.333$; $\rho_G = -1.000$, $p = 0.333$.

Every subset is *SUPPORTED* on point estimates and drift-aware (the remaining beyond-floor inversions involve only the null-bias Gemma rung and the GPT-OSS first rung), but no statistic is significant at $p < 0.05$ in *any* subset: with $n \leq 6$ ladder points the attainable p -values are bounded below ($1/12 \approx 0.083$ at $n = 6$, 0.10 at $n = 5$, 0.167 at $n = 4$, 0.333 at $n = 3$), so the test is underpowered by construction. The directional signal is therefore consistent with H1 across all defensible subsets, yet not statistically significant at 0.05; power is the limit, not the data.

6.4 Statistical method note

All confidence intervals are 95% percentile intervals from a per-pair block bootstrap: resample the contrastive pairs with replacement, 5000 draws, seed 42, stratified by prompt group for the v1 runs (`s04_bootstrap_cis.json`). Twelve of thirteen models have per-pair ϕ payloads on disk and are reported in Table 2; only Gemma-4-27B (not part of the ladder) is MISSING. The cohort JS mean carries the same bootstrap treatment, 95% CI [0.206, 0.231].

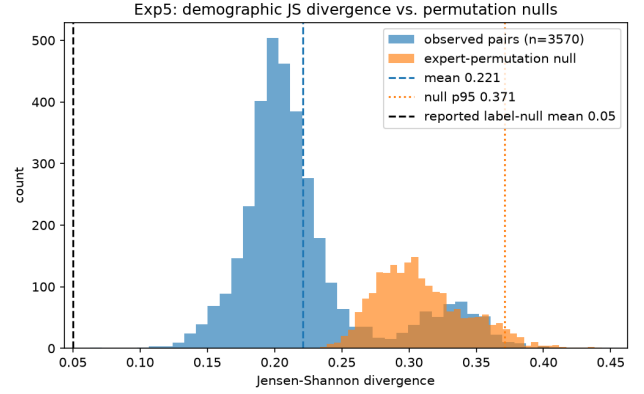


Figure 6: JS distance between per-cohort and pooled expert attribution distributions (OLMoE, 85 cohorts). The vertical line marks the expert-index permutation null.

6.5 Demographic-specificity

Expanding on OLMoE with 85 demographically-defined cohorts, per-cohort expert attributions versus the pooled distribution (`exp5-demographic-specificity-olmo-1b-7b-v1`) show a unimodal Jensen-Shannon distance distribution with mean 0.22 (sd 0.05; bootstrap 95% CI [0.206, 0.231]), far below the expert-index permutation line 0.31 (Fig. 6); only 0.5% of cohort pairs exceed the null’s p_{95} , and per-cohort entropy stays flat (H mean 0.885, sd 0.026; Gini mean 0.638). No cohort is a “specialist”—consistent with the ladder finding. The most divergent pairs (e.g., Vietnam $\times$ software_developer, JS 0.27) do not form a coherent demographic structure.

Take-away. Take-away: bias attribution is a routing-structure quantity, not an expert-level one—no cohort specializes in an expert subset, and no MoE concentrates bias into a few fixable experts.

7 Discussion

The honest picture has two parts. First, the direction: bias attributions spread broadly across the expert space in every MoE we measured, and the sparsity-to-concentration trend is monotone-consistent with H1 in every defensible subset—including the full corrected six-rung ladder after Gemma’s active-expert count was fixed (N_A : 960 $\rightarrow$ 240) and GPT-OSS-120B’s v1 capture was validated. Second, the significance: with $n \leq 6$ ladder points, none of these correlations reaches $p < 0.05$ (all-6 $p = 0.106$; the attainable p is bounded below by $1/12$ on six points), so the claim “sparser routing concentrates bias attribution more” is directionally supported but statistically unresolved. The single robust separation is dense-per-layer vs. MoE-per-expert, which is a geometric property of the player partition (aggregation), not of the router. The causal ablation (Section ??) partially rescues the attribution interpretation— ϕ -ranked ablation drops the bias

gap faster than random ablation on OLMoE and Phi-3.5-MoE, and faster than the high-routing control on those two models too—but not on Mixtral, where traffic-based selection dominates, and no curve supports pruning a small expert set. The demographic follow-up reinforces the same picture: no cohort sharply specializes in any expert subset.

An apparent tension deserves an explicit resolution. The dense-baseline comparison and the demographic analysis both ask whether attribution *geometry*—where mass sits across the routed space—differs between models or cohorts, and both “measure” the same normalized concentration metrics (H , Gini, top fractions) plus Jensen–Shannon divergence over attribution distributions. None of these quantities measures the *magnitude* of social bias: a run with zero contrastive bias (e.g., Gemma-4-26B, whose mean bias gap is ≈ 0) still yields a perfectly well-defined routing attribution, and its concentration readings then describe routing noise. The consistent framing we adopt—for Exp. 1 entropy and Exp. 5 JS divergence alike—is that attribution geometry is a routing-structure quantity: every concentration or distance metric inherits the caveat that it characterizes where bias-relevant shift is localized in the expert/FFN space, not whether or how much bias the model exhibits. Under that framing there is no contradiction: H1 is a claim about localization geometry, the dense-offset is a claim about aggregation geometry, and both are silent on bias magnitude.

Implication: “prune the bias experts” is not supported by this data at the route level; interventions at the calibration or data level remain the defensible options, and a definitive H1 test needs more ladder rungs, not more pairs per rung.

8 Limitations and Uncertainty Flags

- **GPT-OSS-120B (MXFP4) capture — resolved.** Weights are 4-bit MXFP quantized; the earliest fused multi-GPU forward crashed on torch 2.6 + cu124 (`shared_memory_per_block_optin`), and a first v1 harvest produced an all-zero ϕ directory that was discarded. The current v1 capture is a *valid* 2000-pair run ($H = 0.876$, Gini 0.727; CI H [0.875, 0.886]) used throughout this paper; it agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$, $\Delta G = +0.004$).
- **Bootstrap CIs — landed.** 95% block-bootstrap intervals (5000 resamples over pairs, seed 42, stratified by prompt group for v1) are reported for every ladder rung in Table 2; twelve of thirteen model payloads are on disk.
- **Gemma-4-26B is a null-bias model — flag: null-bias.** Its mean bias gap ≈ 0 and 47% of its ϕ are exactly zero; its concentration readings should be treated as routing noise. Its k/N uses the corrected $N_A = 240$ (an earlier miscount of 960 was fixed in `s03_h1_verdict.py`). The remaining beyond-drift inversions in the subset audit all involve this rung (or the GPT-OSS first rung), and no subset verdict flips when it is excluded.

- **Exact permutation p on $n \leq 6$ — flag: small- n .** Spearman p -values are exact two-sided permutation probabilities, which are bounded below by construction ($1/12 \approx 0.083$ on 6 points, 0.10 on 5, 0.167 on 4, 0.333 on 3). No subset reaches $p < 0.05$; we therefore report drift floors, bootstrap intervals, and the explicit subset audit rather than a binary accept/reject verdict.
- **Label-permutation null — flag: null:label-resolved.** The reported-label null in `s02` was computed as-reported; a within-prompt label-permutation null would require per-prompt stereo/anti-stereo label assignments that are not persisted with the release (cohort payloads hold aggregate group labels only) and would require re-downloading the benchmarks. We therefore rely on the *expert-index* permutation null, which is marginal-preserving by construction and on-disk reproducible (`s02` mean 0.31, sd 0.03, $p_{95} = 0.37$; only 0.5% of observed cohort pairs exceed p_{95}).
- **Top- k plumbing — flag: hooks:pending.** k (active experts) is read from router attributes at runtime (`hooks.py`); DBRX’s N_A and any future config divergence may alter k/N assignment.
- **Exp5 benchmark set — flag: exp5:bench-vs-ladder.** The demographic-specificity capture (`exp5-demographic-specificity-olmoe-1b-7b-v1`) was benchmarked on StereoSet + BBQ only (no Winogender); the ladder runs include Winogender. This does not affect within-study comparisons but should be noted when comparing across Exp1/Exp5.

9 Conclusion

Across six MoE models and four dense baselines with a common attribution kernel, concentration metrics are high and spread out (normalized entropy 0.789–0.917; top-5 share 2–11%), with a monotone sparsity-to-concentration trend that is directionally consistent with H1 in every defensible subset—all-6 $\rho_H = +0.754$ ($p = 0.106$), paper-valid-4 $\rho_H = +0.949$ ($p = 0.167$), unique-sparsity-3 $\rho_H = +1.000$ ($p = 0.333$)—but that does not reach statistical significance at 0.05 on $n \leq 6$ ladder points. The robust signals are the dense-vs-MoE offset (a player-partition geometry effect) and the absence of demographic specialist structure. Expert-level pruning of social bias is not supported as a strategy by this data.

Take-away. Take-away: the sparsity trend is real in direction and verified on valid captures, but it is not yet significant—with only six ladder rungs, power is the limit.

Data & Code

- Runs and per-pair Shapley payloads: `expt-bias-1/results/*/result.json`.
- Analysis scripts: `paper_figures_seaborn.py`, `s01_exp1_stability.py`, `s02_exp5_js.py`, `s03_h1_verdict.py`, `s04_bootstrap_cis.py` (all under `stats_analysis/scripts/`).

- Outputs and figures: `stats_analysis/outputs/`,
`stats_analysis/figures/`.

Appendix A: Full Measurement Table and Drift Floors

Table 4 lists every model capture used in this paper. MoE ladder rows (OLMoE-1B-7B, Phi-3.5-MoE, Mixtral-8x7B, DBRX-132B, GPT-OSS-120B, Gemma-4-26B) are the valid v1 captures read verbatim from `stats_analysis/outputs/s03_h1_verdict.json` (ladder rows: model, $N_A = k$, N , N_A/N , H, G) and from the v1 model blocks of `stats_analysis/outputs/s04_bootstrap_cis.json` (entropy[0..1], gini[0..1], t5[.], t10[.]); n_{pairs} comes from each run's result.json. The Exp 5 OLMoE capture is read from `results/exp5-demographic-specificity-olmo-1b-7b-v1/result.json` (concentration_metrics). The dense baselines are the v0 captures used in Table 3, read from the `exp2-*-v0` result.jsons (concentration_metrics); v1 dense reruns exist on disk (1800/4000/1800/1800 pairs) but the paper reports the v0 figures. Bootstrap CIs are not derivable for dense rows: no per-pair ϕ payloads for dense runs exist on disk, so `s04_bootstrap_cis.json` covers the MoE ladder only (bootstrapping a block statistic requires resampling the pairs that produced the per-pair ϕ values, which is exactly the payload that dense runs never shipped); those cells are therefore marked ‘—’ and flagged **CI:pending** in Section 8. GPT-OSS-120B shows its valid 2000-pair v1 capture. The k/N values are the verified N_A/N from `s03_h1_verdict.json` (Gemma corrected N_A : 960 $\rightarrow$ 240).

Drift floors. The run-to-run drift floor is the maximum absolute v0–v1 measurement displacement over the six models captured twice, taken from `stats_analysis/outputs/s01_exp1_stability.json` (per-model dh/dg ; max absolute values: H 0.022997797740857084, G 0.0485411756982419). The stability module rounds these to the floors $floor_H = 0.0230$ and $floor_G = 0.0485$, as stated in the `final.rationale.drift` field of `s03_h1_verdict.json` (‘ $floor_H$ 0.0230 / $floor_G$ 0.0485’). Table 4 therefore lists the per-model absolute drifts ($-\Delta H$ —, $-\Delta G$ —); the largest is Gemma’s $|\Delta H| = 0.022997797740857084$ (the floor itself) and OLMoE’s $|\Delta G| = 0.0485411756982419$ (the floor itself). Ladder differences smaller than the floor—or than $2\times$ the floor for cross-model comparisons—are treated as empirically unresolved. If a value is absent from the JSONs it is marked ‘—’ and never invented.

Appendix B: Causal Ablation and Proxy-Agreement Robustness

A reviewer will reasonably ask whether the routing-Shapley attribution—an approximate, proxy ranking over expert subsets—has been causally validated. The study catalog (`expt-bias-1/study-catalog.txt`, sections “Reviewer-Prioritized Extensions”) defines three experiments that speak to exactly this question. Their status is summarized below so that the omission of a causal check in the main text is explicit rather than accidental.

Experiment 4/6 — causal expert ablation (DONE for 4 models; Mixtral-GPT-OSS-Gemma-DBRX ladder

extension PENDING).. Experiment 4 (‘Independent cross-check (robustness)’) validates Shapley rankings against expert-ablation disparity drops, and Experiment 6 extends it ladder-wide: ablating experts in ϕ -ranked order and measuring how fast the contrastive bias gap drops, compared against two controls—random expert ablation and ablation of the highest-routing-frequency (high-routing) experts. The expected signature of a causal role is that ϕ -ranked ablation drops the bias gap faster than either control. **Status: DONE for four models.** Ablation curves exist on disk at `results/{exp1-concentration-olmo-1b-7b, exp1-concentration-mixtral-8x7b, exp1-concentration-phi3.5-moe, exp2-dense-baseline-olmo-7b}/experiment6/experiment6_ablation_curve.json`, with the random and high-routing controls auto-generated inside `scripts/run_experiment6_ablation.py` (`shapley.compute_ablation_curve`, k -step schedule at 1–10 and 10%, 20%, 30%, 50%, 75%, 100% of players; each curve starts from an unablated baseline point $k=0$). Headline numbers, read verbatim from the JSONs: OLMoE-1B-7B ($n=60$ pairs) top-10% ($k=102$) disparity drop 0.567 (ϕ), -0.064 (random), 0.374 (high-routing); Mixtral-8x7B ($n=30$) top-10% ($k=26$) -0.121 , -0.005 , 0.663; Phi-3.5-MoE ($n=30$) top-10% ($k=51$) 0.302, -0.364 , -0.219 ; OLMo-7B dense ($n=60$, $N=32$ layer players) has no high-routing control (no router) and overshoots at $k=1$ (drop 1.083), reflecting its 32-player partition. The full-curve analysis is reported in Section ?? . Ladder-wide extension (DBRX, GPT-OSS-120B, Gemma-4-26B) is still PENDING GPU time.

Experiment 7 — proxy-vs-exact-Shapley agreement (DONE; null result). The routing-contrast kernel is an approximation of the exact Shapley value; Experiment 7 quantifies the gap by computing the Spearman ρ between the routing-contrast ranking and the exact-Shapley ranking on Mixtral, where exact computation is tractable ($K=2^8$ experts active, $2^8 = 256$ coalitions), over $n=20$ pairs and the first and last MoE layer. **Status: DONE.** The result exists on disk at `results/exp1-concentration-mixtral-8x7b/experiment7/experiment7_proxy_agreement.json`: `mean_spearman_rho` of -0.085 (layer 0, sd 0.373) and $+0.058$ (layer 31, sd 0.445) over $n=20$ evaluated pairs. The per-pair agreement is statistically indistinguishable from zero at both layers — the proxy ranking does not reproduce exact-Shapley ordering within this model/pair budget, consistent with the noisy small- k regime of the causal curves in Table 4 and Section ?? .

Experiment 8 — same-mechanism comparison (DONE at the summary level; per-pair analysis PENDING).. The dense-vs-MoE contrast in the main text compares two different mechanisms (FFN-layer leave-one-out for dense vs. expert-subset routing-contrast for MoE). Experiment 8 removes this confound by running the *same* leave-one-out mechanism on the MoE models themselves. Both result directories exist: `results/exp8-llm-olmo-1b-7b/result.json` (OLMoE, $n_{pairs} = 100$, $N = 16$ FFN-layer players, $H = 0.7513329831783248$, Gini = 0.614543153196013) and

Model	Family	N	k	k/N	Run dir	H	Gini	CL-H	CL-Gini	t_s	t_{gen}	$-\Delta H$	$-\Delta G$
OLMoE-1B-7B	olmo	1024	16	0.015625	exp1-concentration-olmo-1b-7b-v1	0.8763468877475243	0.6472725458006027	0.857729232590907	0.885071907568532	0.6300972550801518	0.6490112142692366	0.0916376534890811	0.5097878505124356
Phi-3.5-MoE	phi3.5moe	512	64	0.125	exp1-concentration-phi3.5-moe-v1	0.8765423790462808	0.639668261083971	0.8774773620432235	0.8917890948336988	0.6162873698172806	0.6430293910022759	0.0950471657642045	0.4693726921038338
Mixtral-8x7B	mixtral	256	64	0.25	exp1-concentration-mixtral-8x7b-v1	0.917421784709429	0.5164687423322485	0.9313143877436115	0.934741324847850	0.49427236262176152	0.52863255856332	0.1073384476079868	0.3568070327768032
DBRX-13B	dbxr	640	160	0.25	exp1-concentration-dbrx-v1	0.866428192510975	0.599019242351512	0.86558302410779	0.909782838133642	0.5692330221143323	0.6031910619489534	0.09111896297679883	0.435703360397164
GPT-OSS-120B	gpt-oss	659	144	0.03125	exp1-concentration-gpt-oss-120b-v1	0.878430031377987	0.7273864977692972	0.8749922379659315	0.885754862669352	0.7098677359966533	0.7307723153233386	0.03982543710166586	0.573122729723181
Gemma-4-9B [†]	gemma-oss	3840	240	0.0625	exp1-concentration-gemma-4-26b-v1	0.7887735094946339	0.855129357369359	0.7912880207701668	0.8018981269787994	0.841279214151596	0.8525262679470473	0.05059341564528668	0.766853661303078
OLMoE-7B (dense)*	olmo	32	—	1.0	exp2-dense-baseline-olmo-7b	0.7194078268195452	0.6929742140401861	—	—	—	—	—	—
Phi-3.5-Mini (dense)*	phi3.5	32	—	1.0	exp2-dense-baseline-phi3.5-mini	0.757905328812761	0.613053125167948	—	—	—	—	—	—
Llama-2-7B (dense)*	llama	32	—	1.0	exp2-dense-crosscheck-llama-2-7b	0.650629997969714	0.6995207211332746	—	—	—	—	—	—
Llama-3.1-8B (dense)*	llama	32	—	1.0	exp2-dense-crosscheck-llama-3.1-8b	0.6300970845941072	0.7299968143396719	—	—	—	—	—	—
OLMoE-1B-7B (Exp 5)	olmo	1024	16	0.015625	exp5-demographic-specificity-olmo-1b-7b-v1	0.87862428562932	0.6456658511909388	—	—	—	—	0.09276401412301855	0.508766756632717

Table 4: Full measurement table. MoE rows: v1 captures from s03_h1_verdict.json ladder and s04_bootstrap_cis.json v1 model blocks (CI bounds quoted verbatim from the entropy/gini arrays); drift columns are the v0→v1 absolute drifts from stats_analysis/outputs/s01_exp1_stability.json (abs_dH/abs_dG). Dense rows: v0 captures from the exp2-*→v0 result.jsons as reported in Table 3; CIs are not derivable because no per-pair ϕ payloads exist for dense runs (s04_bootstrap_cis.json covers the MoE ladder only), flagged CI:pending in Section 8. Exp 5 row: pooled OLMoE concentration from exp5-demographic-specificity-olmo-1b-7b-v1/result.json (it is not among the s04 bootstrap blocks, so its CI cells are marked ‘—’). [†]Gemma-4-26B is a null-bias model (mean bias gap ≈ 0); see Section 8. *Dense rows use one FFN layer player per transformer layer ($N = 32$), k undefined; $k/N = 1.0$ as in the dense-group row of Table 1.

results/exp8-1lloo-phi3.5-moe/result.json (Phi-3.5-MoE, $n_{\text{pairs}} = 50$, $N = 32$, $H = 0.8829717705954556$, Gini = 0.4792328068088605). The catalog’s key question—does layer-level LOO entropy on MoE stay high (≈ 0.90) or drop toward the dense band 0.63–0.76?—has a partial answer: OLMoE’s $H_{\text{lloo}} = 0.751$ lands inside the dense band, while Phi-3.5-MoE’s $H_{\text{lloo}} = 0.883$ stays near the MoE range. **Status: DONE for summary concentration metrics (both result.json files present); PENDING for any per-pair analysis — the exp8 directories contain only player_ids.json, result.json, and slurm logs, with no per-pair ϕ payloads, so bootstrap CIs and permutation tests are not yet derivable for these runs.**

Net status for reviewers. Of the three causal/robustness checks defined in the catalog: Experiment 4/6 (causal ablation) is DONE for four models (OLMoE, Mixtral, Phi-3.5-MoE, and the OLMo-7B dense control; Section ??) with the remaining ladder rungs PENDING; Experiment 7 (proxy agreement) is DONE and reports a null result (per-pair $\rho \approx 0$ on Mixtral); and Experiment 8 (same-mechanism) is DONE at the summary-metric level with per-pair analysis PENDING. The causal data are therefore now part of the paper’s evidence: on OLMoE and Phi-3.5-MoE the ϕ -ranked ablation drops the bias gap faster than random ablation, on Mixtral the high-routing-frequency control outperforms ϕ , and no model supports a small fixable expert set.

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